

Attention Kernel Portability Across Six GPUs and Two Inference Regimes: A Measurement Study

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Abstract

We study whether an attention backend selected on one GPU remains a good choice on another. We benchmark forward prefill and single-token KV-cache decode on six NVIDIA GPUs with fixed library versions. A backend is the separated winner of a configuration when the runner-up is at least 10% slower and the lower bound of a 95% bootstrap interval on the latency ratio exceeds 1. This holds in 128 of 447 forward-prefill and 409 of 1,203 decode configurations. Among separated comparisons across Ampere, Ada and Hopper GPUs, the winner changes in 27 of 64 prefill comparisons and 29 of 289 decode comparisons. Reusing the source GPU’s choice on the target costs a median $1.002\times$ for prefill and $1.000\times$ for decode, with 95th percentiles of $1.610\times$ and $1.325\times$. In some cases the source’s choice cannot run on the target at all, and we report those separately. Each GPU was measured on its own host, so the results do not separate architecture from host effects.

1 Introduction

A practitioner choosing an attention kernel usually reads a benchmark run on a single GPU. Does the winner of that benchmark stay the winner on a different GPU?

Serving systems usually assume it does. They fix an attention backend by a preference order and use that order on every device they deploy to. If the ranking is a property of the kernel, that is fine. If it is a property of the kernel and device together, an order tuned on one GPU may be wrong on another.

We time 10 prefill and 6 decode implementations over 403 configurations on the six GPUs in Table 1. The 73,230 attempts give 11,835 usable (GPU, configuration, implementation) medians.¹

¹Code: <https://github.com/Srivatsan6923/attention-kernel-portability>. The measurements

Every claim compares a ranking within one device against the ranking within another. We never claim that one device is faster than another.

The study makes two contributions. The first is a measurement protocol that records which kernel actually ran and names a winner only when it is separated from the runner-up. The second is the finding that a backend choice holds only for the GPU, regime and library versions it was measured on.

2 Background and related work

FlashAttention (Dao et al., 2022) keeps the $N \times N$ score matrix out of HBM and is evaluated on A100. FlashAttention-2 (Dao, 2024) is also evaluated on A100, and FlashAttention-3 (Shah et al., 2024) targets H100 only. Each reports results on the GPU its authors measured, which is the practice this study tests. Flash-Decoding (Dao et al., 2023) adds split-KV for single-token queries on A100, and FlashInfer (Ye et al., 2025) builds it into a serving engine evaluated on A100 and H100.

Compiler and kernel benchmarks follow the same pattern. The PyTorch 2 paper (Ansel et al., 2024) evaluates TorchInductor, which generates Triton (Tillet et al., 2019), on one A100. KernelBench (Ouyang et al., 2025) profiles 250 PyTorch workloads on an L40S, and TritonBench (Li et al., 2025) evaluates 184 Triton operators on an A100. Serving systems such as vLLM (Kwon et al., 2023), Sarathi-Serve (Agrawal et al., 2024) and DistServe (Zhong et al., 2024) are evaluated on A100, and Splitwise (Patel et al., 2024) on A100 and H100. TVM (Chen et al., 2018) and Ansor (Zheng et al., 2020) instead search for an implementation per target, which treats the best implementation as a property of the hardware. In HPC the question is well studied. Williams et al.

are in release <https://github.com/Srivatsan6923/attention-kernel-portability/releases/tag/v1.2> with SHA-256 checksums.

(2009) give per-architecture performance ceilings, and Pennycook et al. (2016) and Deakin et al. (2019) define performance portability over a stated set of platforms.

3 Study design

Implementations and grid. The prefill set is a naive PyTorch reference, a `torch.compile` path in three variants, the three PyTorch SDPA backends (memory-efficient, flash and cuDNN), the Triton fused-attention tutorial kernel with a small BF16 change, and FlashAttention 2 and 3. The decode set is a naive KV-cache implementation, an Inductor path, SDPA, `flash_attn_with_kvcache`, FlashInfer, and FlashAttention prefill at query length 1. A *cell* is one configuration, defined by regime, batch, query and KV head counts, head dimension, sequence length, dtype, forward or forward-plus-backward, launch mode, cache state and causality. The full grid has 159 prefill and 244 decode cells. Table 1 gives per-device coverage.

Timing protocol. Backend order is randomised per configuration, and each backend’s timing blocks run back to back. Because backends are not interleaved, we do not assume host or thermal effects cancel within a comparison. In eager mode, each block’s CUDA-event time is divided by its call count. We take the median over blocks within a process launch, then the median over launches. Blocks target about 200 μ s. For cold-cache cells, L2 is flushed once before each block, so only the first call in a block is fully cold. CUDA-graph cells are timed with Triton’s `do_bench_cudagraph`. Confidence intervals come from a 2000-draw bootstrap that resamples whole process launches, because calls within a launch share clock and thermal state. The median between-process coefficient of variation (CV) is 0.011 on A10 (n=1871) and 0.026 on H100 (n=2474).

Processes are pinned to the NUMA node closest to the GPU. In two runs of the H100 decode grid that differ only in pinning, the median between-process CV over 966 matched (cell, implementation) pairs falls from 0.112 unpinned to 0.034 pinned, and the share above 25% CV falls from 30.7% to 9.9%. These numbers use block-level timing and are not directly comparable with the per-call CVs above.

Definitions. Let $t_{g,c,b,r}$ be the median latency of an attention call in process launch r , for GPU g ,

configuration c and backend b . The reported latency is the median over launches,

$$T_{g,c,b} = \text{median}_r t_{g,c,b,r}. \quad (1)$$

With b_1 and b_2 the fastest and second-fastest eligible backends at (g, c) , the separation ratio and rule are

$$R_{g,c} = \frac{T_{g,c,b_2}}{T_{g,c,b_1}}, \quad (2)$$

$$\text{Sep}(g, c) = \mathbf{1}[R_{g,c} \geq 1.10 \wedge L_{95}(R_{g,c}) > 1], \quad (3)$$

where L_{95} is the lower end of a 95% bootstrap interval on the same ratio. Both conditions are required. Failing the rule does not show that the backends are equal. It means no fastest backend could be established. In all 2,082 cells used here, the fastest and second-fastest backends ran in the same set of launches. In the wider scan over every implementation pair, 50 of 29,304 pairs, all on the RTX 5090 and all involving the naive reference, lost a launch to an out-of-memory failure. Those pairs are compared on the launches both backends share.

Let $\mathcal{B}_{g,c}$ be the backends with a usable measurement at (g, c) and $b_{g,c}^*$ the winner. Let $\mathcal{E}$ be the set of matched comparisons (g, h, c) where c was measured on both GPUs and $\text{Sep}(g, c) = \text{Sep}(h, c) = 1$. The winner-change rate is

$$\text{WCR} = \frac{\sum_{(g,h,c) \in \mathcal{E}} \mathbf{1}[b_{g,c}^* \neq b_{h,c}^*]}{|\mathcal{E}|}. \quad (4)$$

The common-backend variant limits each comparison to $\mathcal{B}_{g,c} \cap \mathcal{B}_{h,c}$ for that configuration and recomputes b^* and Sep inside that set.

The cost of reusing GPU g ’s choice on GPU h is

$$C_{g \rightarrow h, c} = \frac{T_{h,c,b_{g,c}^*}}{\min_{b \in \mathcal{B}_{h,c}} T_{h,c,b}}. \quad (5)$$

A value of 1.20 means the source’s choice is 20% slower than the fastest backend on the target. When $b_{g,c}^* \notin \mathcal{B}_{h,c}$ the ratio is undefined. We count those cases and never substitute the target’s own best.

Correctness gate. Every implementation is checked against a reference, and only rows that pass enter a ranking. Of 73,230 attempts, 54,213 (74.0%) completed with status OK. Of those, 54,098 passed the check and are eligible. The other 115 belong to one class that was never checked on its own device and are excluded. Of

GPU	cc	family	BW (GB/s)	planned repeats pre/dec	cells pre/dec
A100	8.0	Ampere	1732.8	5/5	144/192
A10	8.6	Ampere	483.3	5/5	144/191
L40	8.9	Ada	644.4	3/3	144/192
L40S	8.9	Ada	644.4	5/5	144/192
H100	9.0	Hopper	2979.7	5/5	159/244
RTX 5090	12.0	Blackwell	1517.5	5/5	144/192

Table 1: GPUs and coverage. Bandwidth comes from a copy benchmark on each host. *Planned repeats* are separate process launches per configuration, for prefill and decode. L40 kept a median of two usable repeats. *Cells* counts configurations with at least one eligible measurement.

the remaining attempts, 16,326 were UNSUPPORTED, 1,590 were skipped by an out-of-memory predictor, 950 raised a run-time error (all FlashInfer on the RTX 5090, Section 6), and 151 ran out of memory despite passing the predictor. No correctness check failed. Bandwidth is measured on each host with a copy benchmark, not taken from a datasheet.

4 Dispatch validation

A latency says something about a kernel only if that kernel actually ran. We tried to record the launched kernel names for every timed row and got a readable trace for 36,054 of the 54,213 OK attempts (66.5%). Grouping rows by the fields that decide dispatch (implementation, GPU, head dimension, dtype, GQA ratio, mode and launch) gives 651 dispatch classes, and 625 of them (96.0%) have at least one traced row. A class confirms the kernel family only. Within a class the exact kernel can change with batch, for example between `flash_fwd_splitkv_combine` and `flash_fwd`, but the family is always the one requested.

Among traced rows, the requested kernel family is missing from the trace in 162 cases, a mismatch rate of 0.449%. Over all OK rows that is 0.299%. We treat the untraced 33.5% as unknown, not as matching. The mismatches cluster, with 66 of the 162 coming from the SDPA memory-efficient backend on H100.

For TorchInductor, 3,645 of 17,537 Inductor rows recorded the `fuse_attention` counter, and it was 0 on every one of them across all six GPUs. Inductor’s attention-fusion pattern therefore never fired on those rows. The counter does not tell the whole story, though. 45 traced Inductor rows on H100, from 6 distinct cells, did launch a flash kernel.

5 Portability

Most configurations have no clear winner.

Only 28.6% of forward-prefill and 34.0% of decode configurations have a separated winner (128 of 447 and 409 of 1,203). Figure 1 gives the rate per GPU. A flip between two backends within a few percent of each other is not a measured difference, so Table 2 reports every rate over all shared configurations and over those separated on both GPUs.

Prefill winners change far more often than decode winners.

Table 2 pools the ten pairs among A10, A100, L40, L40S and H100. Over all shared configurations the two regimes look similar (66.0% and 52.1%). Keeping only separated comparisons pulls them apart. Prefill changes winner in 27 of 64 comparisons (42.2%) and decode in 29 of 289 (10.0%). Restricting each pair to the backends both GPUs ran, and recomputing winners in that set, gives 33 of 72 for prefill and 27 of 289 for decode. This restriction removes backends that only one GPU could run, such as FlashAttention 3, and it does not lower the prefill rate. Figure 2 shows what reusing a choice costs. The median reused choice is $1.002\times$ the target’s best in prefill and $1.000\times$ in decode, but the 95th percentiles are $1.61\times$ and $1.32\times$.

The RTX 5090. The lower half of Table 2 repeats the calculation for the five pairs that include the RTX 5090. Decode changes winner in 63 of 85 comparisons and prefill in 16 of 30. In most of the decode cases the other GPU’s winner is FlashInfer, which produced no timing on the 5090 (Section 6). We did not measure what FlashInfer would have cost there, so we do not attribute the high rate to architecture.

6 Backend availability

FlashInfer produced no timing on the RTX 5090.

FlashInfer is the decode winner on 128 of 191 cells on A10, 133 of 192 on A100, 131 of 244 on H100 and 87 of 192 on L40, but only 49 of 192 on L40S, where SDPA leads. On the RTX 5090 it never ran. Of 960 attempts, 950 raised `RuntimeError: FlashInfer requires GPUs with sm75 or higher` across 190 cells, and the other 10 ran out of memory. The message names sm75 on an sm120 GPU, so it does not explain the failure. Without FlashInfer, the RTX 5090’s decode winner is `flash_attn_with_kvcache` on 137 of 192 cells,

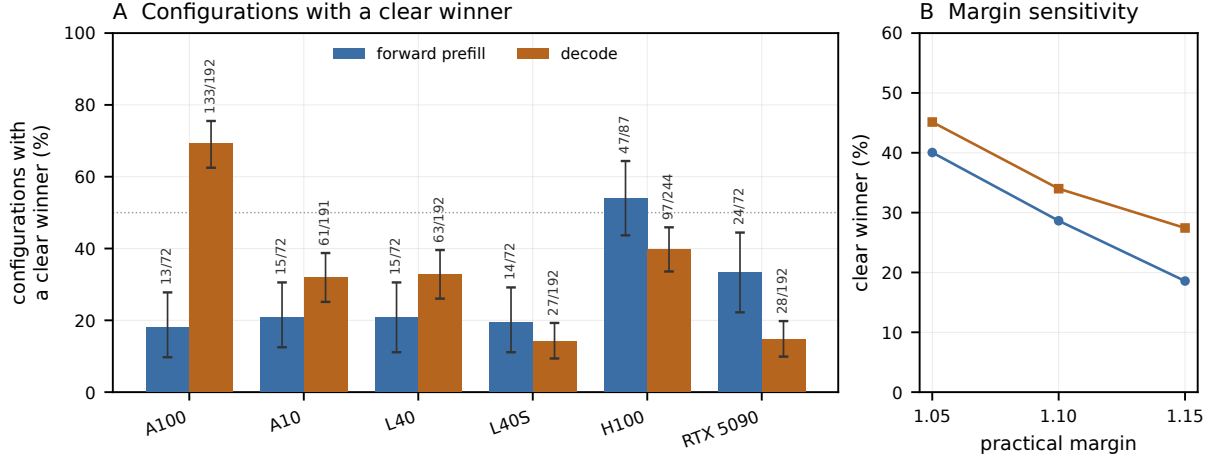


Figure 1: Share of configurations with a clear (separated) winner, by GPU. Whiskers are 95% intervals, and the counts above each bar are separated over total configurations. Workload counts differ by GPU. Panel B shows the pooled rate at margins of 1.05, 1.10 and 1.15.

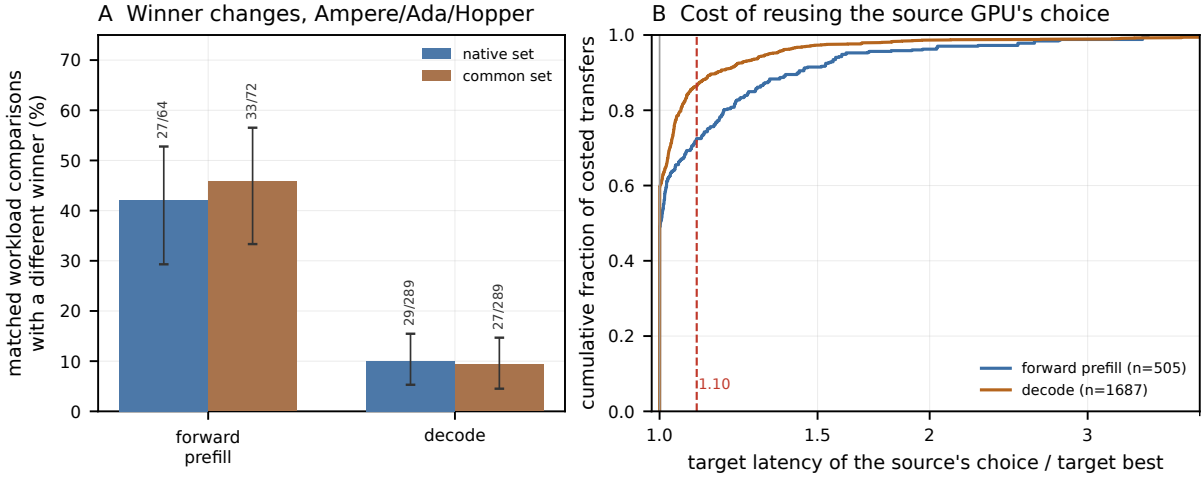


Figure 2: Left, the winner-change rate over the ten Ampere, Ada and Hopper GPU pairs, using each GPU’s own backends (native set) and only the backends both GPUs ran (common set). Right, the distribution of reuse cost over all ordered pairs of the six GPUs where the source has a separated winner. In 85 of 590 prefill and 287 of 1,974 decode cases the source’s choice had no usable timing on the target, and these are not plotted. The cost distribution includes cases where the winner did not change.

at a median of 115.7 μ s.

Library builds. Table 4 lists which GPU targets each installed library was built for. Over forward prefill, FlashAttention 2 wins 35 of 72 cells on A100 and 13 of 72 on the RTX 5090, both of which have native code, but 2 of 72 on L40 and none on A10 or L40S, which run A100 code under binary compatibility. On H100 it wins none of 87, and FlashAttention 3 wins 48. The pattern matches the packaging, but we did not rebuild flash-attn for the missing targets, so we do not claim the packaging causes it. FlashInfer compiles its kernels at run time, so its coverage depends on the host toolkit. Every other library in the audit ships native sm_120 code built with the

same CUDA 12.8, so the toolkit can target sm_120 and the FlashInfer failure remains unexplained.

7 Threats to validity

One host per GPU. Each GPU was measured on the host that had it, so device differences are mixed with host differences. A100 prefill was re-measured on a second host because the first run kept too few repeats. Every ranking is formed within one GPU and one regime, so no comparison mixes the two hosts, and their copy bandwidths agree within 0.05%. A10 and A100 decode ran a few days before the other runs, on an earlier harness version whose timing code is identical.

device set	regime	all shared cells		separated both	
		n	flip	n	flip [95% CI]
Amp/Ada/Hop	prefill	720	66.0%	64	42.2% [29, 53]
	decode	1916	52.1%	289	10.0% [5, 15]
5090 vs rest	prefill	360	65.8%	30	53.3% [30, 74]
	decode	959	78.4%	85	74.1% [56, 90]

Table 2: Winner-change rate pooled over GPU pairs. *Amp/Ada/Hop* is the ten pairs among A10, A100, L40, L40S and H100, and *5090 vs rest* is the five pairs with the RTX 5090. *n* is the number of pair comparisons and *flip* the share with a different winner. The right columns keep only configurations separated on both GPUs. Intervals are a 2000-draw cluster bootstrap over configurations, since one configuration appears in several pairs.

Small separated samples. Requiring a separated winner on both GPUs leaves 64 of 720 prefill and 289 of 1,916 decode comparisons among Ampere, Ada and Hopper GPUs. This is the right filter for a ranking claim, but it means the headline rates rest on tens to a few hundred comparisons. One of the 11,835 usable measurements has a single launch and cannot be separated. L40 has fewer repeats than the other GPUs, and Appendix A shows the headline rates barely move without it.

Untraced rows. 33.5% of OK attempts have no kernel trace, and 26 of 651 dispatch classes have no traced row. The 0.449% mismatch rate applies to traced rows only.

Modelled quantities. Effective KV bandwidth is an analytic byte count divided by measured latency, not measured DRAM traffic, so its ratio to the copy benchmark does not show saturation.

What was not measured. The hosts we used did not allow Nsight Compute or Nsight Systems, so we make no occupancy, cache or counter-based roofline claims. We tested one combination of torch, flash-attn and FlashInfer, and a different version is a different experiment. We time a single attention call, not end-to-end serving latency, on GPUs from one vendor.

8 Conclusion

In the tested configurations, the fastest forward-prefill backend changed across Ampere, Ada and Hopper GPUs much more often than the fastest decode backend. The median cost of reusing a choice was small, but the upper tail and the cases where the choice could not run at all mean the median is not enough to pick a backend. Backend

choices should be measured on the GPU and workload they will serve, with uncertainty and coverage reported.

A Repeat-count sensitivity

L40 had three planned repeats instead of five and kept a median of two, so its intervals are the widest. Dropping it moves the separation rate from 28.6% to 30.1% for forward prefill and from 34.0% to 34.2% for decode.

B Backend availability

Table 3 lists which comparisons were possible.

C Library builds and GPU support

The audit shows which GPU targets the installed libraries were built for. It shows what could run, not why anything was fast. CUDA lets code built for a lower minor compute capability run on a higher one within the same major version.

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backend	regime	GQA	support	OK	unsup.	fail
D0-naive-kv	dec	both	YYYYYY	5530	0	10
D1-inductor	dec	expanded	YYYYYY	194	5346	0
D2-sdpa	dec	native	YYYYYY	5535	0	5
D3-fa-kvcache	dec	native	YYYYYY	5515	0	25
D4-flashinfer	dec	native	YYYYYY	4565	0	975
D6-fa-prefill-at-1	dec	native	YYYYYY	5515	0	25
P0-naive	pre	both	YYYYYY	3148	0	851
P1-inductor	pre	both	YYYYYY	3209	0	790
P1-inductor-nofuse	pre	both	YYYYYY	121	3873	5
P1-inductor-where	pre	both	YYYYYY	121	3873	5
P2b-sdpa-mem-eff	pre	native	YYYYYY	3969	30	0
P2c-sdpa-flash	pre	native	YYYYYY	3999	0	0
P2d-sdpa-cudnn	pre	native	YYYYYY	3999	0	0
P3-triton	pre	both	YYYYYY	3999	0	0
P4-fa2	pre	native	YYYYYY	3999	0	0
P4h-fa3	pre	native	..Y...	795	3204	0

Table 3: Which comparisons are possible at all. *support* is one character per device in the order A10, A100, H100, L40, L40S, RTX 5090: Y produced at least one usable row there, . attempted and produced none, - never attempted. *unsup.* counts configurations the backend declined in advance and *fail* runtime errors plus allocations refused or hit. FlashInfer reads YYYYYY.: it ran on five devices and produced no timing on the RTX 5090. FlashAttention 3 reads ..Y..., H100 only, by construction.

library	sm_80	sm_86	sm_89	sm_90	sm_120	PTX
flash-attn 2.8.3	SASS	80	80	SASS	SASS	none
libtorch_cuda	SASS	SASS	SASS	SASS	SASS	none
cuDNN engines	SASS	SASS	86	SASS	SASS	sm_120
cuBLAS	SASS	SASS	86	SASS	SASS	sm_120
flashinfer 0.6.18	JIT	JIT	JIT	JIT	none	n/a
flash-attn 3	n/f	n/f	n/f	n/f	n/f	n/f

Table 4: Binary targets of the installed libraries, from cuobjdump. SASS means native code is present. An italic number means no native code shipped and the GPU runs code built for that older architecture. PTX is embedded PTX the driver can compile. JIT means the library builds kernels at run time, *none* means no kernel was produced for that target, and *n/f* means the audit did not find the library. FlashAttention 3 did run on H100, so its n/f row is a gap in the audit.

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